Testing Plan: Smart Content Summarizer

1. Purpose

This testing plan verifies that SummaSmart accepts user instructions and an article URL, retrieves webpage content, generates an instruction-based summary, and renders a complete Markdown report.

2. Test Environment

- Application: Google Opal
- Workflow: Smart Content Summarizer (SummaSmart)
- Test interface: Opal Preview
- Retrieval tool: Connected Get Webpage tool
- Output: Rendered Markdown webpage

3. Test Cases

TC-001: User Instructions Input

Objective: Verify that the application collects summarization requirements.

Input: "Summarize the webpage in exactly 3 bullet points using simple English."

Expected Result: The instructions are accepted and passed to the summary generation step.

Observed Result: The Preview requested summarization instructions.

Status: PASS

TC-002: Article URL Input

Objective: Verify that the application accepts a webpage URL.

Input: `https://example.com`

Expected Result: The URL is accepted and passed to the retrieval step.

Observed Result: The Preview accepted the example.com URL and proceeded through the workflow.

Status: PASS

TC-003: Webpage Retrieval

Objective: Verify that the connected retrieval tool obtains article content.

Input: `https://example.com`

Expected Result: The webpage's text is available to the summary generation step.

Observed Result: The final output displayed the Example Domain heading and webpage text.

Status: PASS

TC-004: Summary Generation

Objective: Verify that the summary reflects the retrieved article.

Input: Retrieved Example Domain content and user instructions.

Expected Result: A concise summary based on the retrieved content.

Observed Result: The output contained three bullet points about the Example Domain website.

Status: PASS

TC-005: Output Formatting

Objective: Verify that the requested bullet count is followed.

Input: "Summarize the webpage in exactly 3 bullet points using simple English."

Expected Result: Exactly three bullet points.

Observed Result: The example.com output displayed exactly three bullet points.

Status: PASS

TC-006: Source URL Display

Objective: Verify that the original source URL is visible and clickable.

Expected Result: The final page contains a clickable link to the original article.

Observed Result: The final page displayed `https://example.com` as a clickable link.

Status: PASS

TC-007: User Instructions Display

Objective: Verify that the final page displays the instructions used to generate the summary.

Expected Result: A clearly labeled User Instructions section.

Observed Result: The final page displayed the submitted instructions.

Status: PASS

TC-008: Original Web Content Display

Objective: Verify that the retrieved content appears separately from the summary.

Expected Result: An Original Web Content section containing retrieved text.

Observed Result: The final page displayed the Example Domain heading and original webpage text separately.

Status: PASS

TC-009: Longer Article Retrieval

Objective: Check whether the application can retrieve and summarize a longer webpage.

Input: `https://en.wikipedia.org/wiki/Artificial_intelligence`

Expected Result: The article is retrieved and summarized according to the user instructions.

Observed Result: The Preview displayed "Oops, something went wrong" during the test.

Status: FAIL — Investigation required.

Notes: Successful retrieval of example.com does not establish successful retrieval of Wikipedia. Check the connected retrieval tool and the target webpage's accessibility before concluding the cause.

TC-010: Retrieval Error Handling

Objective: Verify that retrieval failures do not produce fabricated summaries.

Input: A URL that the retrieval tool cannot access.

Expected Result: A clear retrieval error message and no invented summary.

Observed Result: Not yet verified with a controlled retrieval-failure test.

Status: PENDING

4. Test Summary

Test Area	Result
User instructions	PASS
URL input	PASS
Example webpage retrieval	PASS
Summary generation	PASS
Three-bullet formatting	PASS
Source URL display	PASS
Instructions display	PASS
Original content display	PASS
Longer article retrieval	FAIL
Retrieval error handling	FAIL

5. Known Issues

1. Wikipedia retrieval failed during Preview testing.
2. The cause of the retrieval failure has not yet been confirmed.
3. Retrieval error handling requires a dedicated test.
4. Advanced dashboard interactions depend on the capabilities of the Opal output renderer.

6. Exit Criteria

The project is ready for final submission when:

- The five workflow steps are connected correctly.
- The successful example test is repeatable.
- A second accessible article is tested.
- Retrieval error handling is tested and documented.
- The final Markdown output contains all required sections.
- All repository files are uploaded and the explanation video is reviewed.

7. Testing Record

Record the date, test input, expected result, observed result, and final status whenever a test is run again. Update this document to reflect actual results rather than marking unverified tests as passed.